

Motor Control Board

Variant: DRAFT

2026-08-24

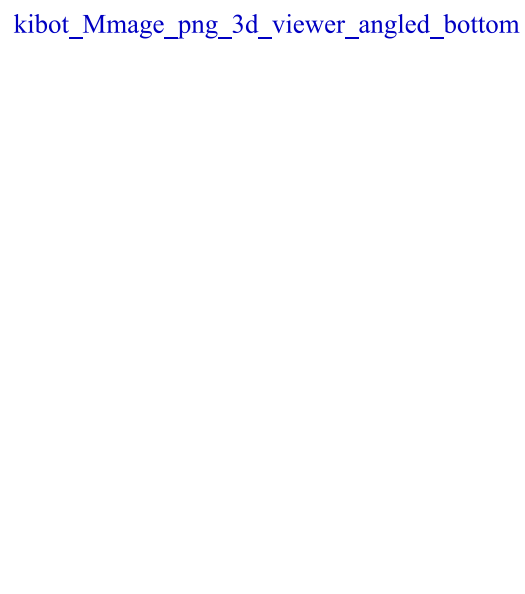
Rev + (Unreleased)

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TOP VIEW



BOTTOM VIEW



DESIGN CONSIDERATIONS

DESIGN NOTE:

Example text for informational design notes.

DESIGN NOTE:

Example text for debug notes.

DESIGN NOTE:

Example text for cautionary design notes.

DESIGN NOTE:

Example text for critical design notes.

LAYOUT NOTE:

Example text for critical layout guidelines.

NOTES

Comment

Not fitted components are marked as **X**

- DRAFT - Very early stage of schematic, ignore details.
- PRELIMINARY - Close to final schematic.
- CHECKED - There shouldn't be any mistakes. Contact the engineer if you find any.
- RELEASED - A board with this schematic has been sent to production.

Date: 24-Aug-2026

Block Diagram

File: Block Diagram.kicad_sch

Revision History

File: Revision History.kicad_sch

Project Architecture

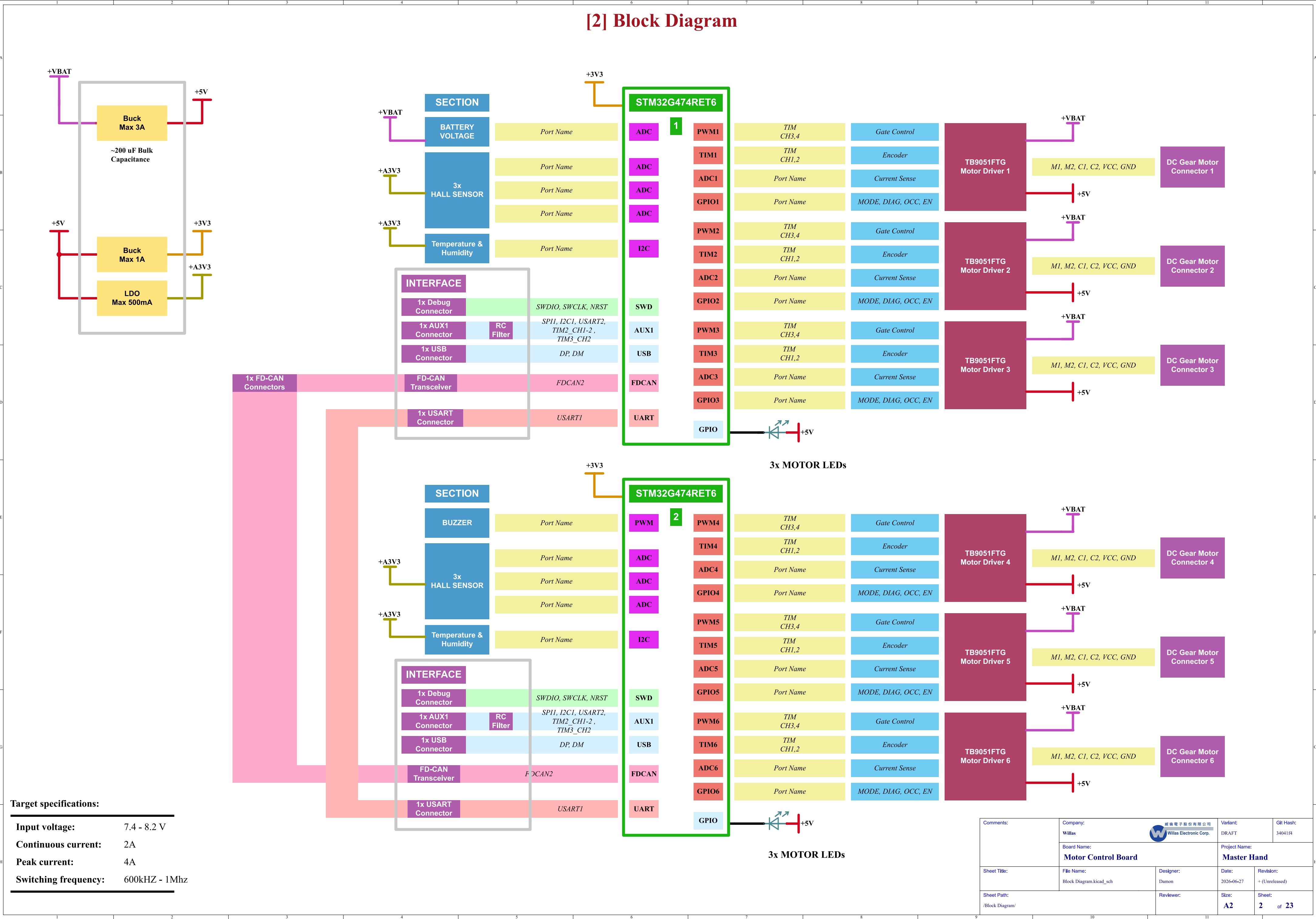
File: Project Architecture.kicad_sch

Power - Sequencing

File: Power - Sequencing.kicad_sch

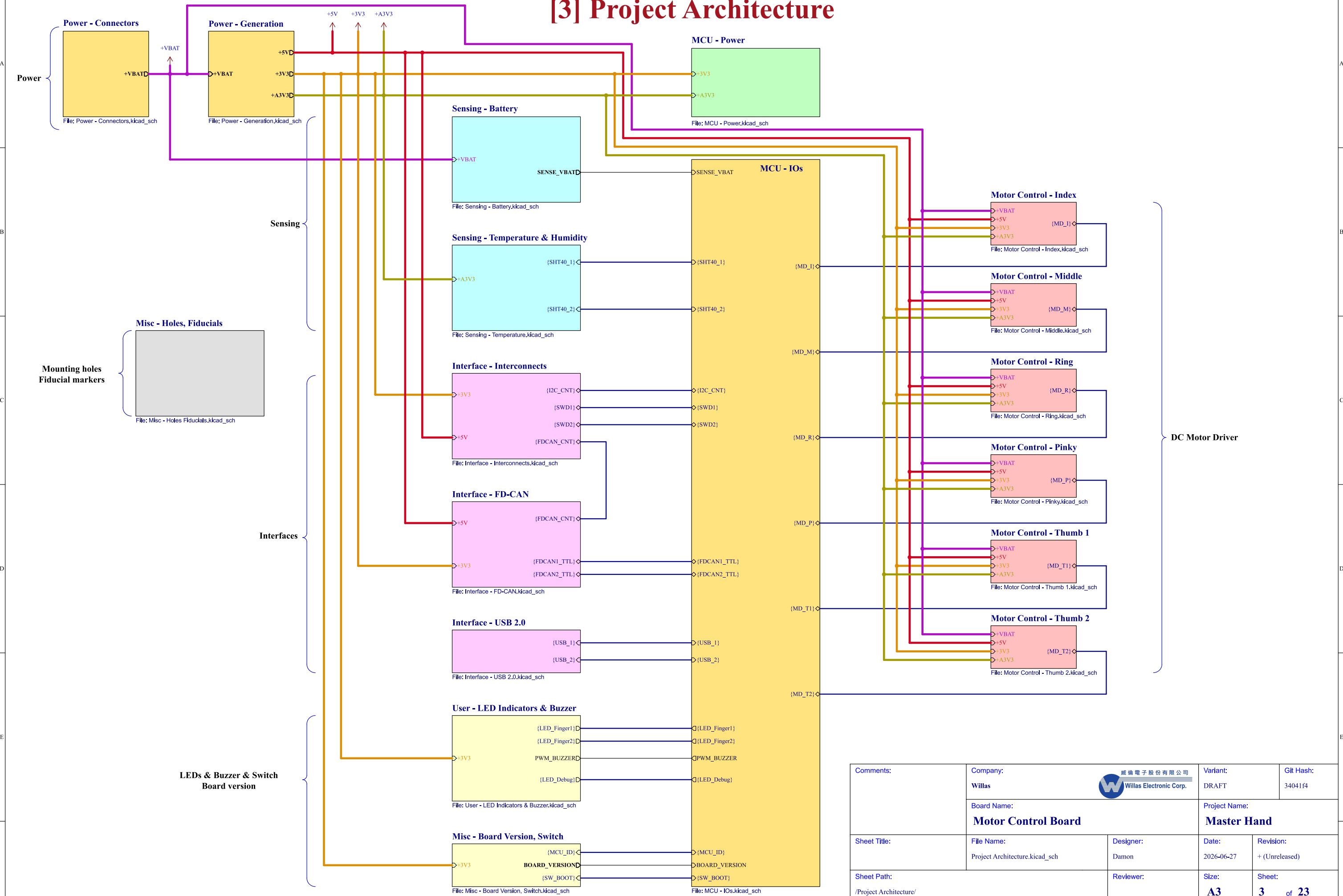
Comments:	Company: Willas		 威倫電子股份有限公司 Willas Electronic Corp.		Variant: DRAFT	Git Hash: 34041f4
	Board Name: Motor Control Board				Project Name: Master Hand	
Sheet Title:	File Name: Master_Hand_Kicad.kicad_sch		Designer: Damon	Date: 2026-06-27	Revision: + (Unreleased)	
Sheet Path: /			Reviewer:	Size: A3	Sheet: 1 of 23	

[2] Block Diagram



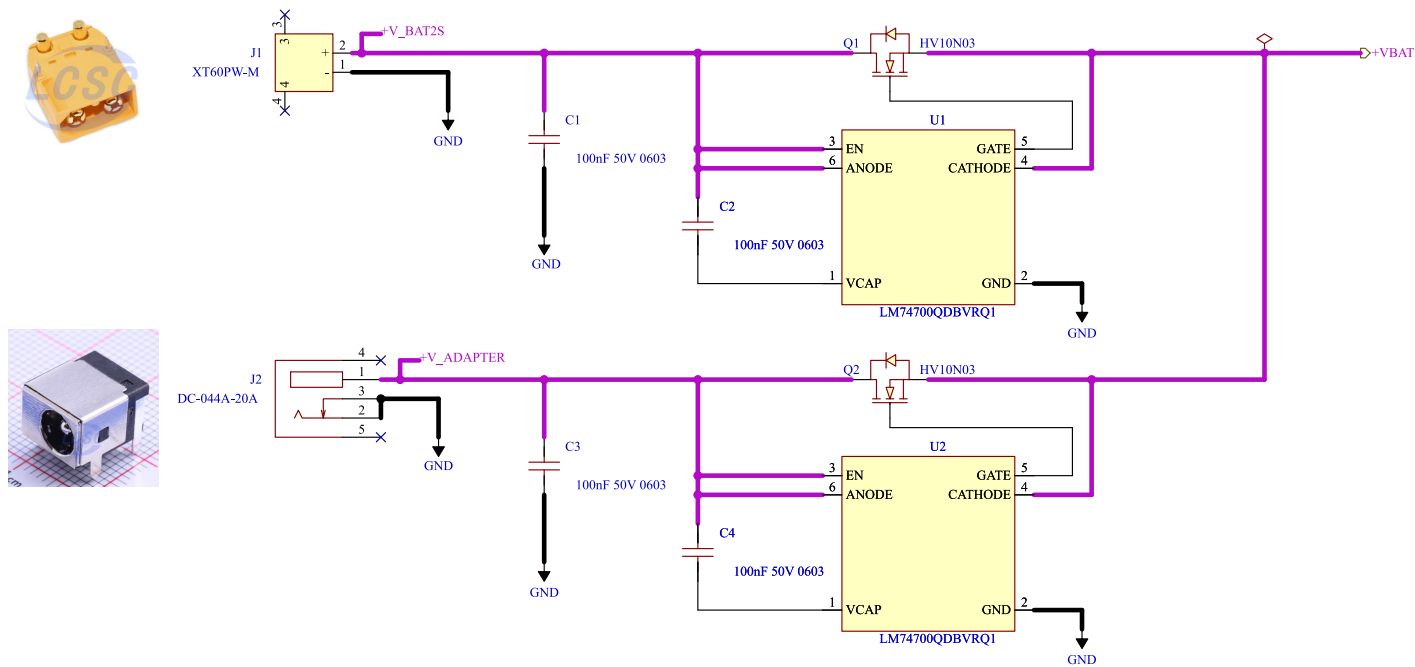
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	Board Name: Motor Control Board		Project Name: Master Hand	
Sheet Title: Block Diagram.kicad_sch	File Name: Block Diagram.kicad_sch	Designer: Damon	Date: 2026-06-27	Revision: + (Unreleased)
Sheet Path: /Block Diagram/	Reviewer:		Size: A2	Sheet: 2 of 23

[3] Project Architecture



Comments:	Company: Willas		 Willas Electronic Corp.	Variant: DRAFT	Git Hash: 34041f4
	Board Name: Motor Control Board			Project Name: Master Hand	
Sheet Title:	File Name: Project Architecture.kicad_sch	Designer: Damon	Date: 2026-06-27	Revision: + (Unreleased)	
Sheet Path: /Project Architecture/		Reviewer:	Size: A3	Sheet: 3 of 23	

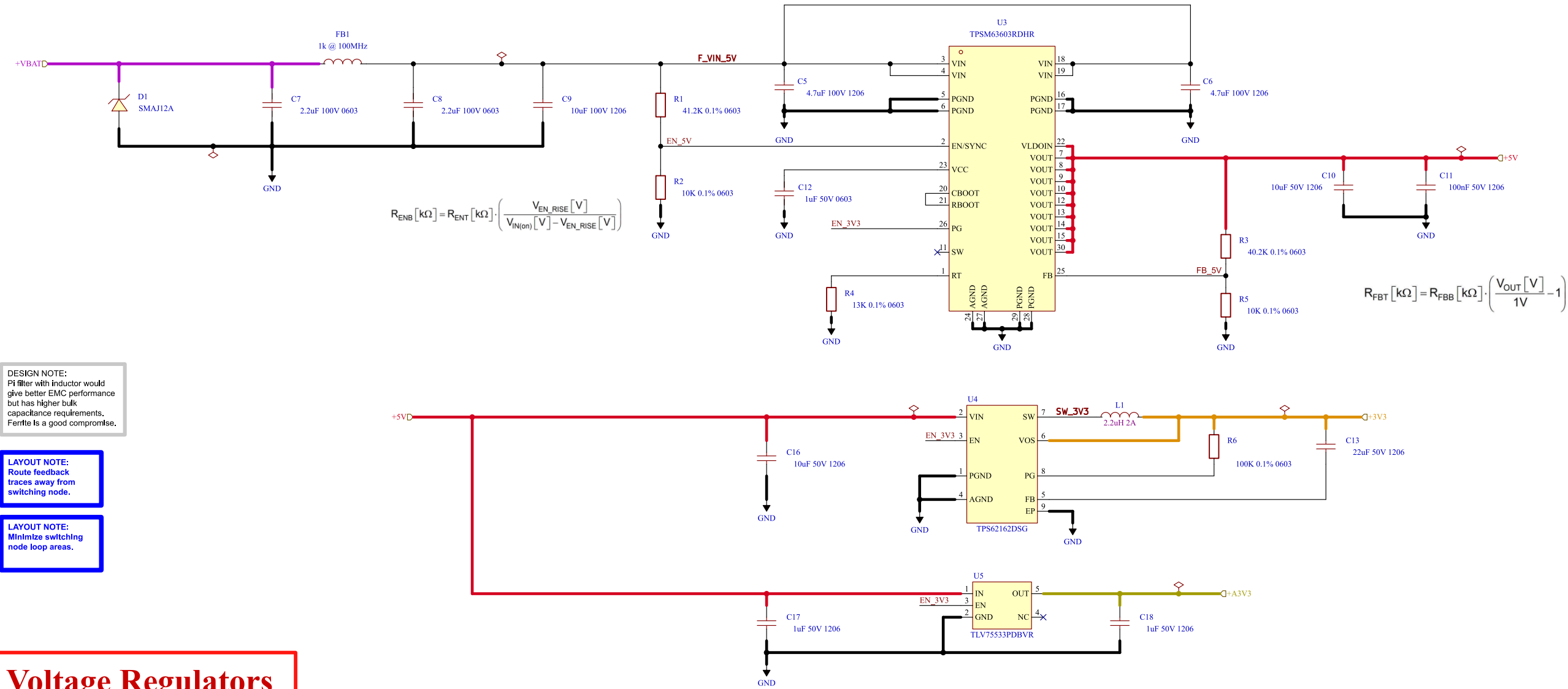
[4] Power - Connectors



Power Connectors - Path Control

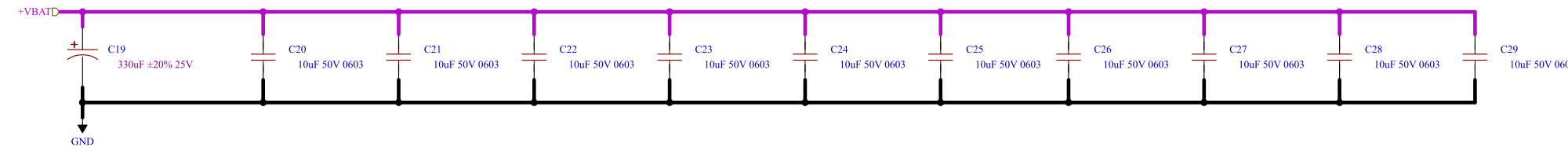
Comments:	Company: 威倫電子股份有限公司 Willas		Variant: DRAFT	Git Hash: 34041f4
	Board Name: Motor Control Board		Project Name: Master Hand	
Sheet Title:	File Name: Power - Connectors.kicad_sch	Designer: Damon	Date: 2026-06-27	Revision: + (Unreleased)
Sheet Path: /Project Architecture/Power - Connectors/		Reviewer:	Size: A4	Sheet: 4 of 23

[5] Power - Generation



Voltage Regulators

DC BULK



DC-Link Bulk Capacitors. 330uF + 10 x 10 uF = 430uF (35V)

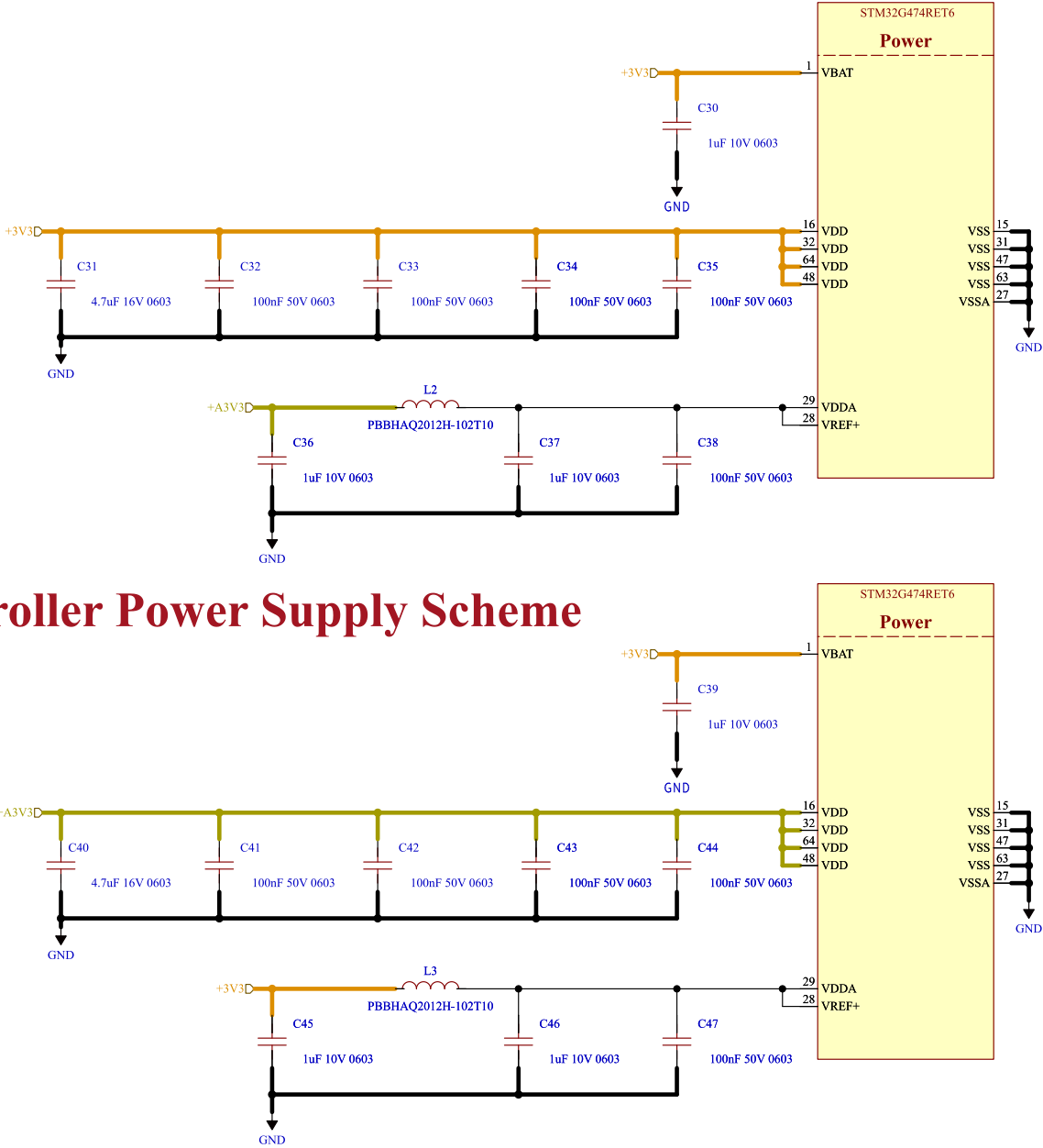
Test Points



Comments:	Company: Willas		 Willas Electronic Corp.	Variant: DRAFT	Git Hash: 34041f4
	Board Name: Motor Control Board			Project Name: Master Hand	
Sheet Title:	File Name: Power - Generation.kicad_sch		Designer: Damon	Date: 2026-06-27	Revision: + (Unreleased)
Sheet Path: /Project Architecture/Power - Generation/			Reviewer:	Size: A3	Sheet: 5 of 23

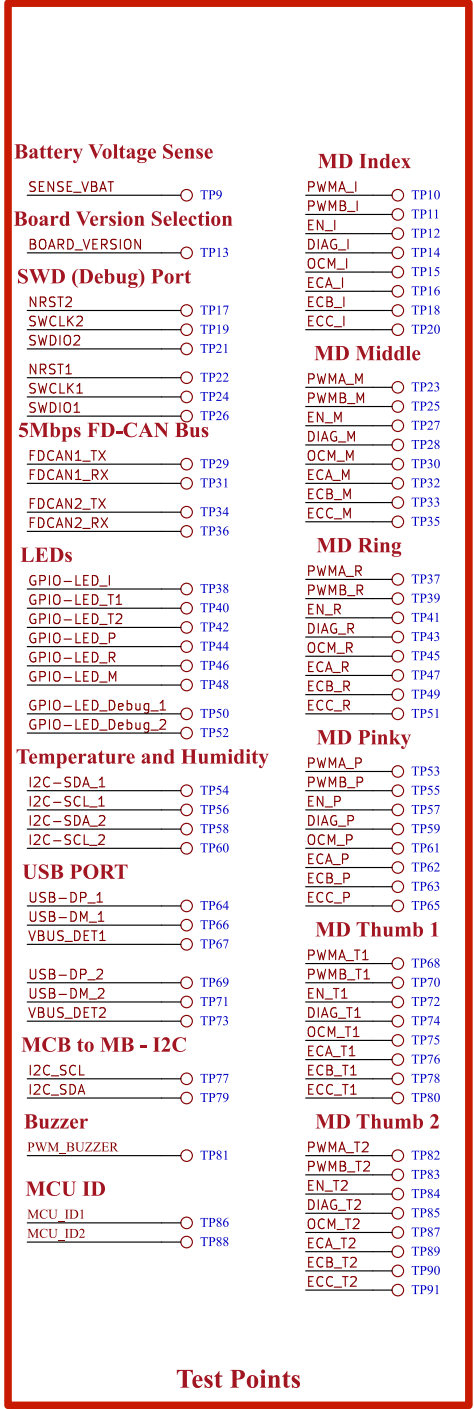
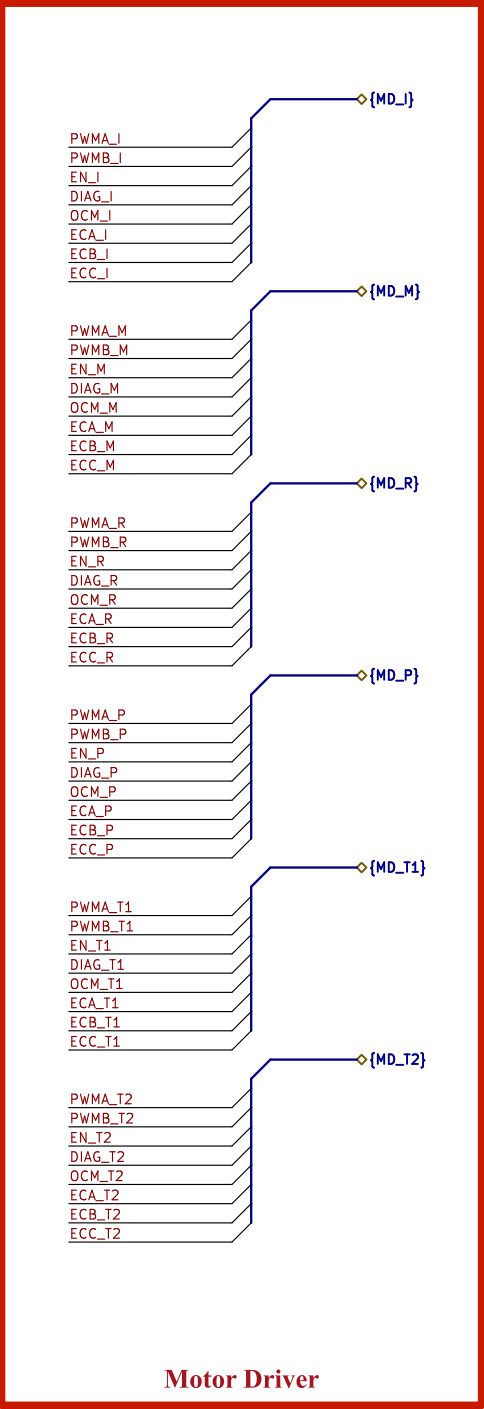
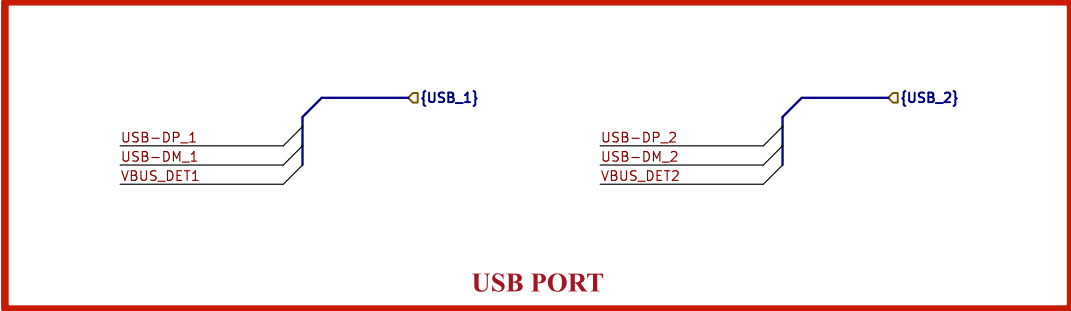
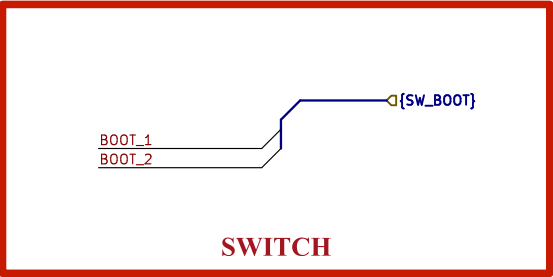
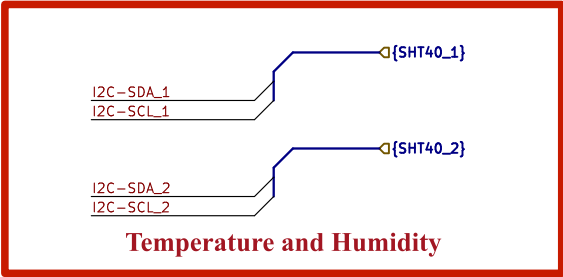
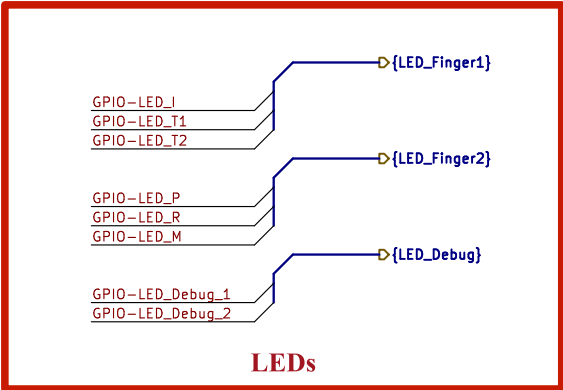
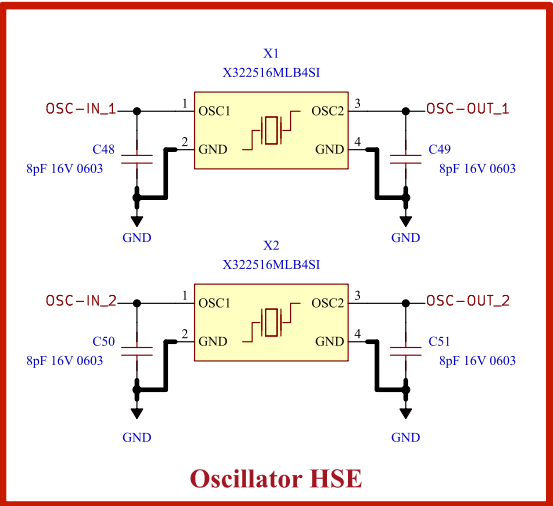
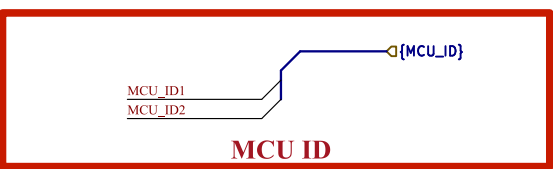
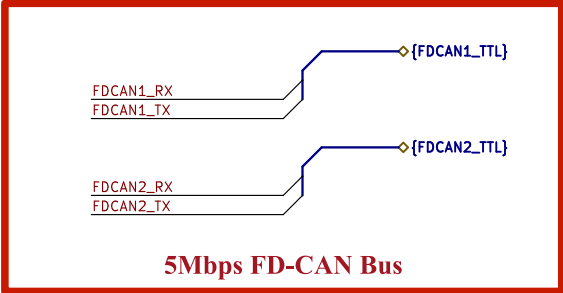
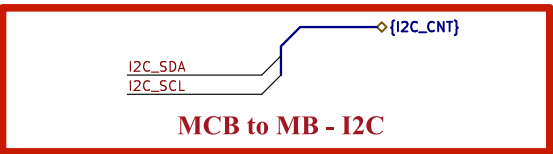
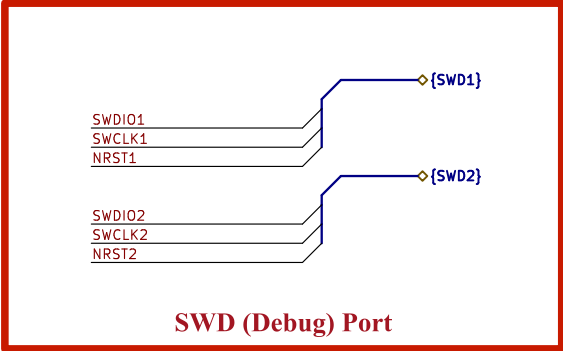
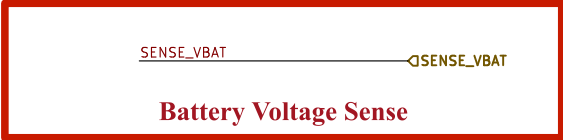
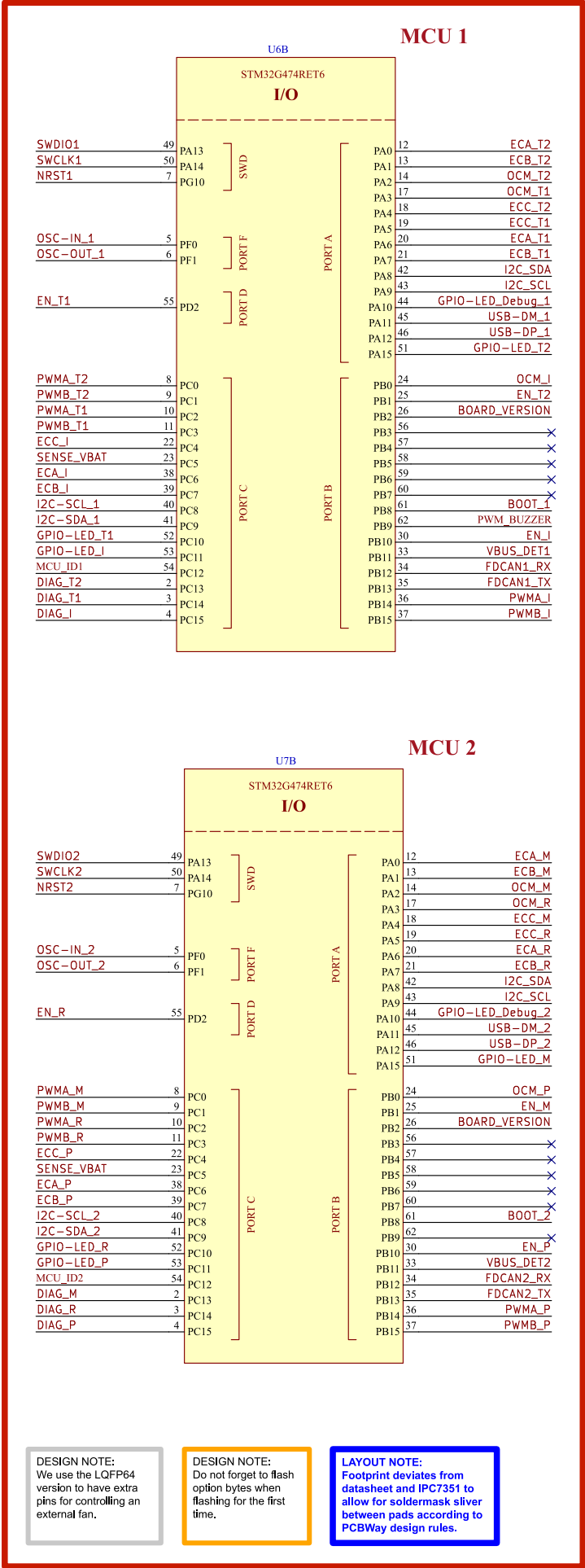
Microcontroller Power Supply Scheme

LAYOUT NOTE:
Place each capacitor close
to the relevant pin.
Minimize loop inductance.



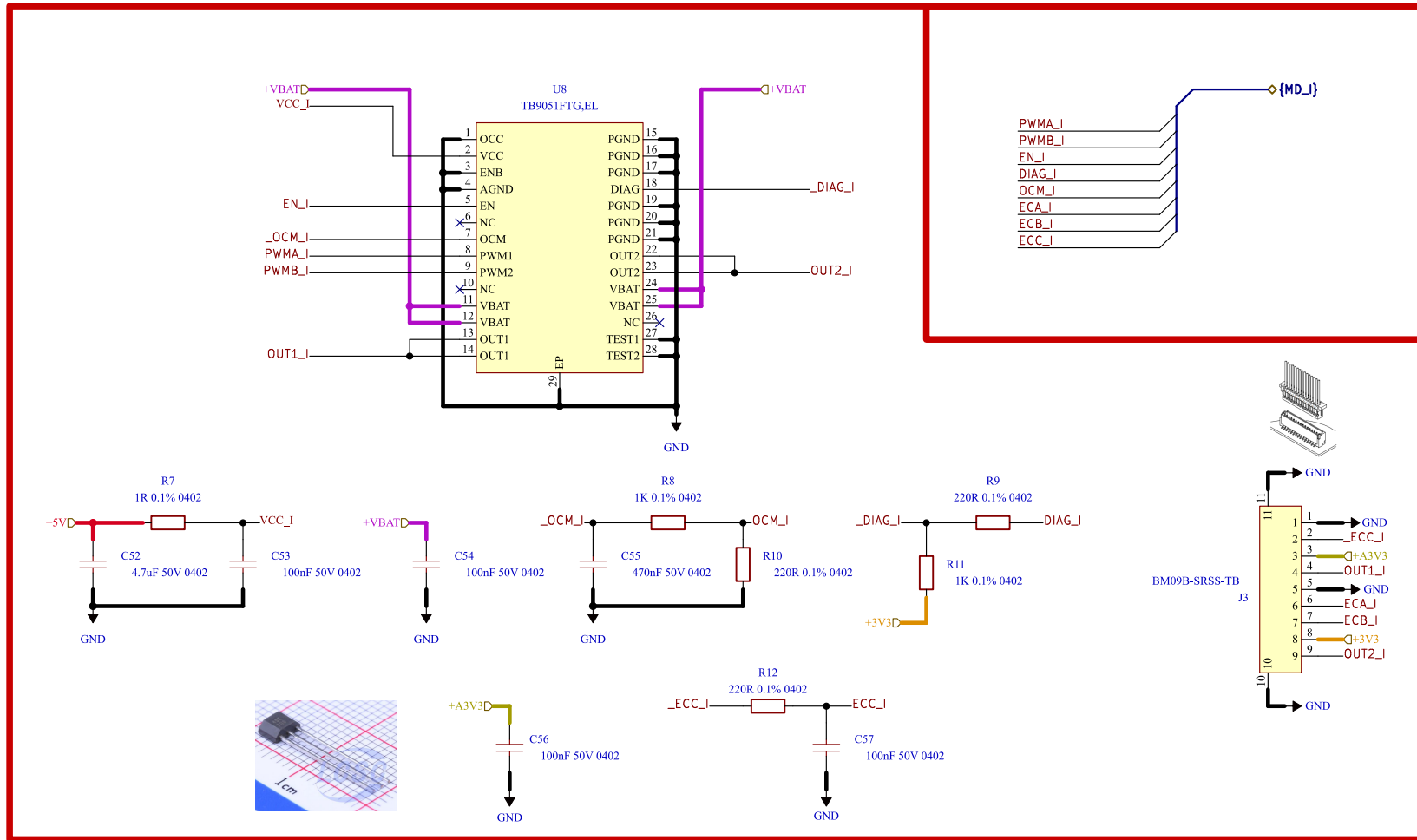
Comments:	Company: Willas		 威倫電子股份有限公司 Willas Electronic Corp.		Variant: DRAFT	Git Hash: 34041f4
	Board Name: Motor Control Board			Project Name: Master Hand		
Sheet Title:	File Name: MCU - Power.kicad_sch	Designer: Damon	Date: 2026-06-27	Revision: + (Unreleased)		
Sheet Path: /Project Architecture/MCU - Power/		Reviewer:	Size: A3	Sheet: 6 of 23		

[7] MCU - IOs



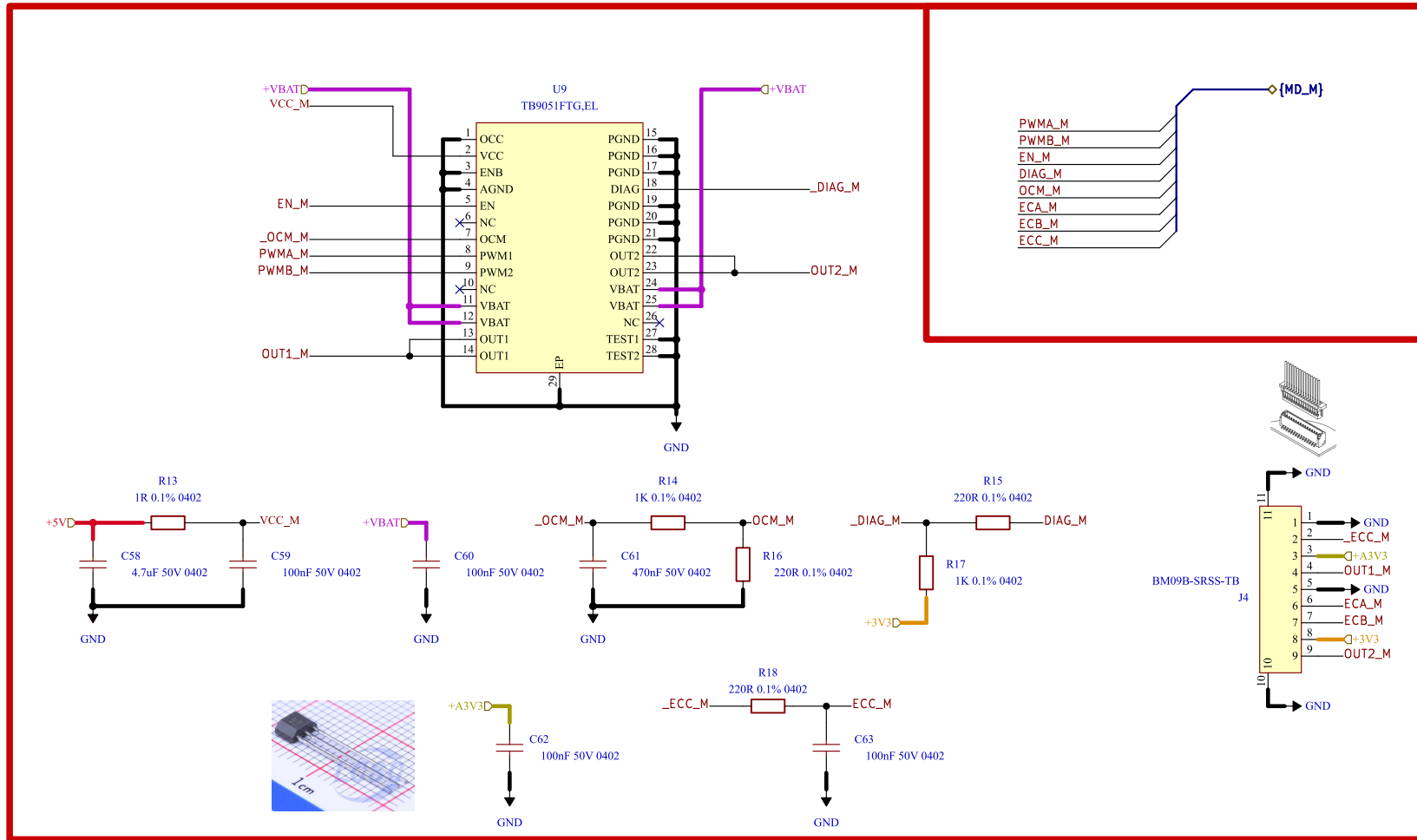
Comments:	Company: Willas		 威倫電子股份有限公司 Willas Electronic Corp.	Variant: DRAFT	Git Hash: 34041f4
	Board Name: Motor Control Board			Project Name: Master Hand	
Sheet Title:	File Name: MCU - IOs.kicad_sch	Designer: Damon	Date: 2026-06-27	Revision: + (Unreleased)	
Sheet Path: /Project Architecture/MCU - IOs/		Reviewer:	Size: A3	Sheet: 7 of 23	

[8] Motor Control - Index



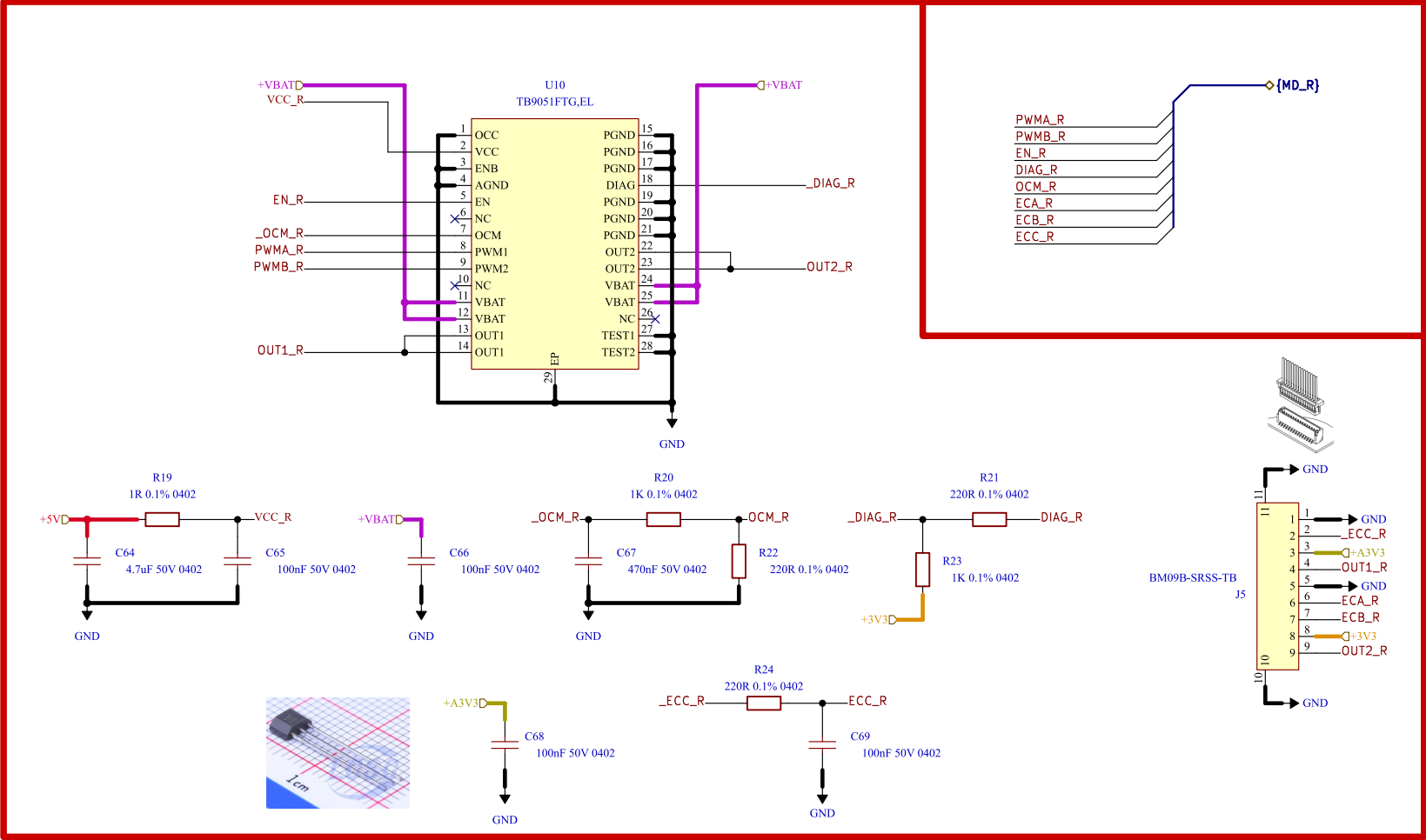
Comments:	Company: 威倫電子股份有限公司 Willas		Variant: DRAFT	Git Hash: 34041f4
	Board Name: Motor Control Board		Project Name: Master Hand	
Sheet Title:	File Name: Motor Control - Index.kicad_sch	Designer: Damon	Date: 2026-06-27	Revision: + (Unreleased)
Sheet Path: /Project Architecture/Motor Control - Index/		Reviewer:	Size: A4	Sheet: 8 of 23

[9] Motor Control - Middle



Comments:	Company: 威倫電子股份有限公司 Willas		Variant: DRAFT	Git Hash: 34041f4
	Board Name: Motor Control Board		Project Name: Master Hand	
Sheet Title:	File Name: Motor Control - Middle.kicad_sch	Designer: Damon	Date: 2026-06-27	Revision: + (Unreleased)
Sheet Path: /Project Architecture/Motor Control - Middle/		Reviewer:	Size: A4	Sheet: 9 of 23

[10] Motor Control - Ring



Comments:

Company:

Willas



Variant:

DRAFT

Git Hash:

34041f4

Board Name:

Motor Control Board

Project Name:

Master Hand

Sheet Title:

File Name:

Motor Control - Ring.kicad_sch

Designer:

Damon

Date:

2026-06-27

Revision:

+(Unreleased)

Sheet Path:

/Project Architecture/Motor Control - Ring/

Reviewer:

Size:

A4

Sheet:

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[11] Motor Control - Pinky

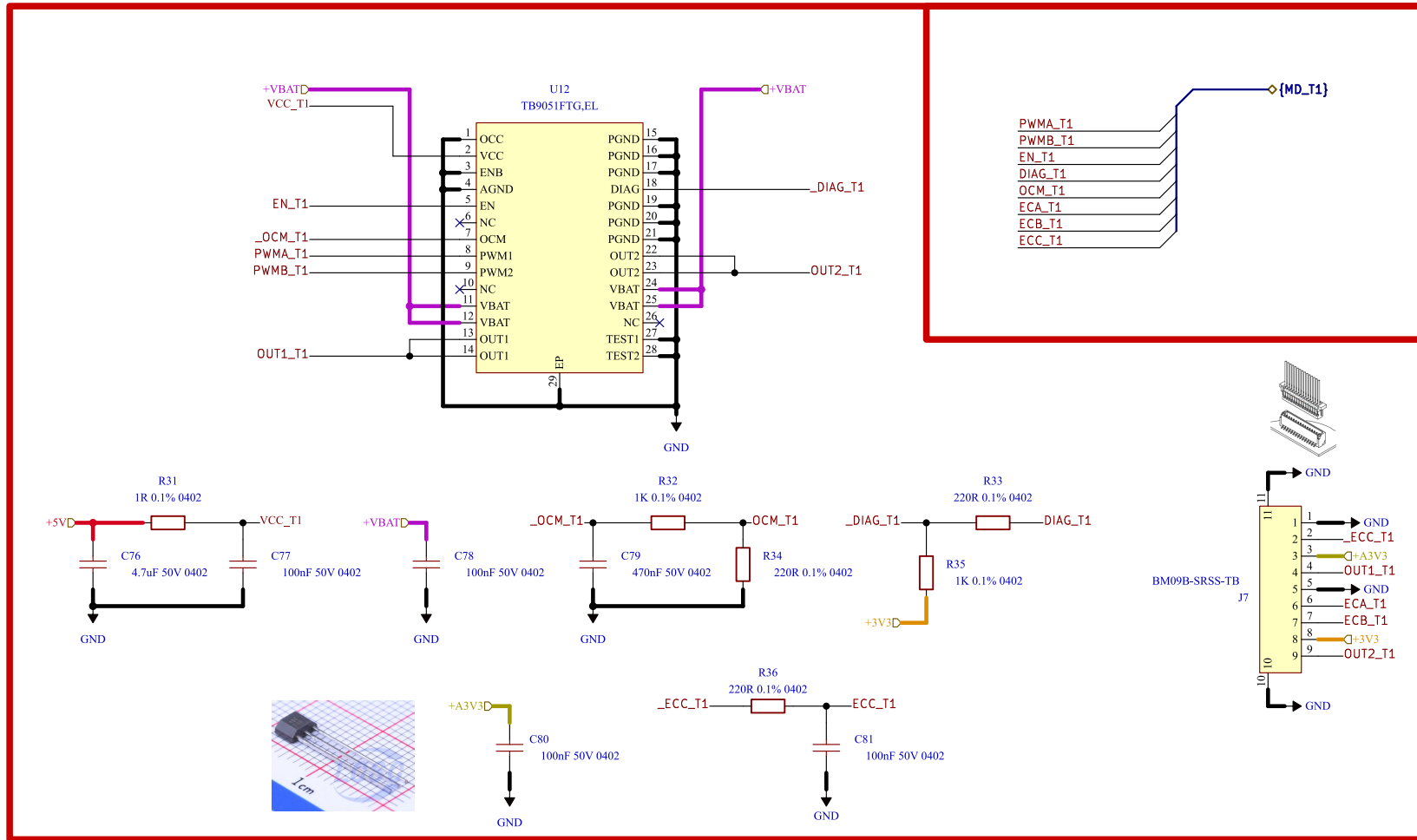
The schematic diagram illustrates the Motor Control Board (Pinky) circuit. The central component is the TB9051FTG.EL motor driver IC (U11). The pin connections are as follows:

- Power and Ground:** +VBAT (VCC_P) to pin 1, GND to pin 29. +VBAT to pin 11, GND to pin 26.
- Control Signals:** EN_P to pin 3, _OCM_P to pin 8, PWM1_P to pin 9, PWM2_P to pin 10, OUT1_P to pin 13, OUT2_P to pin 23, _DIAG_P to pin 18, and MD_P to pin 24.
- Motor Signals:** PMA_P, PWM_P, EN_P, DIAG_P, OCM_P, ECA_P, ECB_P, and ECC_P are connected to the motor driver pins.
- Resistors and Capacitors:** R25 (1R 0.1% 0402), R26 (1K 0.1% 0402), R27 (220R 0.1% 0402), R28 (220R 0.1% 0402), R29 (1K 0.1% 0402), R30 (220R 0.1% 0402), C70 (4.7uF 50V 0402), C71 (100nF 50V 0402), C72 (100nF 50V 0402), C73 (470nF 50V 0402), C74 (100nF 50V 0402), and C75 (100nF 50V 0402).
- Connector:** BM09B-SRSS-TB (J6) with pins 1-9 connected to GND, +A3V3, OUT1_P, ECA_P, ECB_P, +3V3, and OUT2_P.

Comments:	Company:	Variant:	Git Hash:
	Willas	DRAFT	34041f4
	Board Name:	Project Name:	
	Motor Control Board	Master Hand	
Sheet Title:	File Name:	Designer:	Date:
	Motor Control - Pinky.kicad_sch	Damon	2026-06-27
Sheet Path:	Reviewer:	Size:	Revision:
/Project Architecture/Motor Control - Pinky/		A4	+ (Unreleased)
			Sheet:
			11 of 23

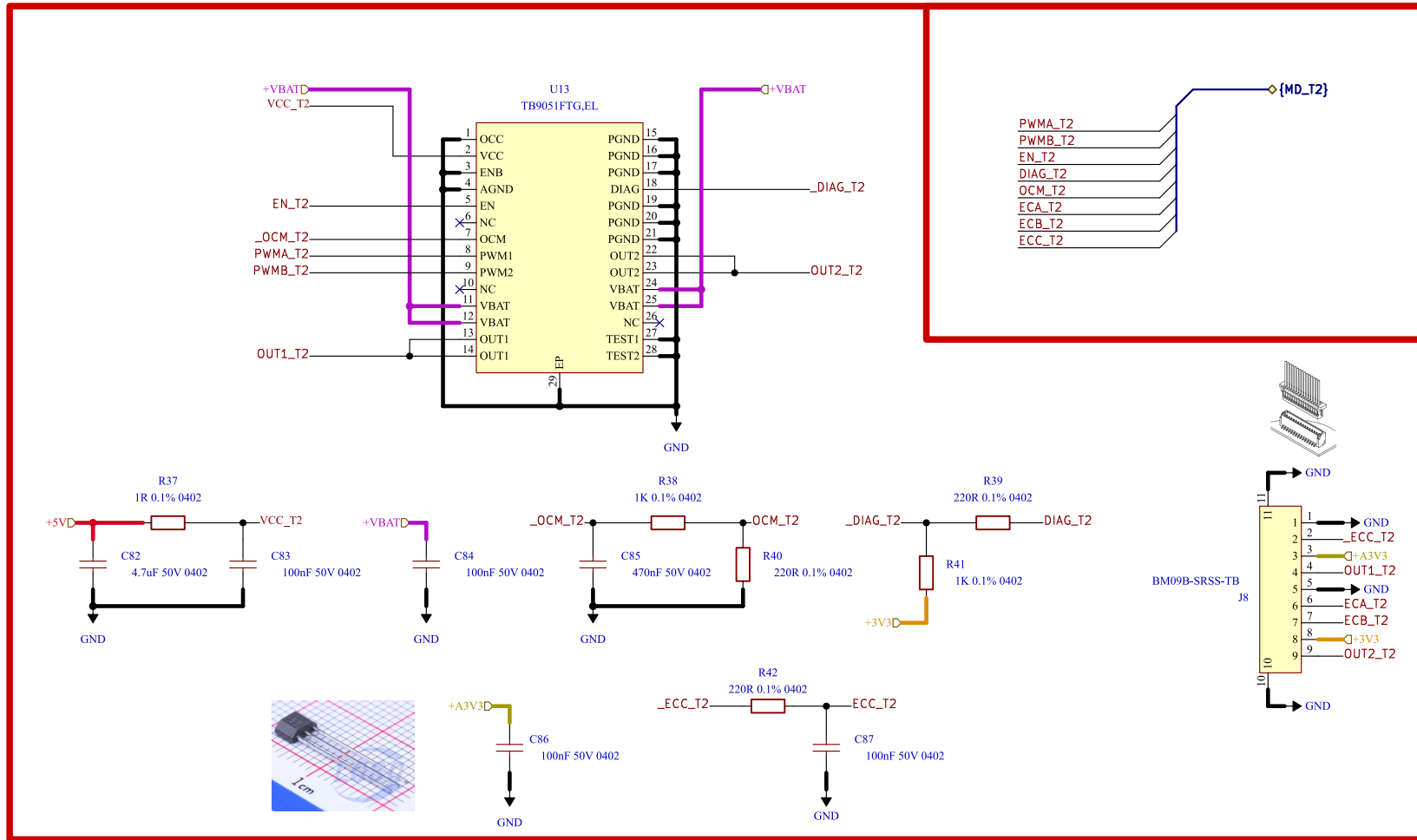
Comments:	Company: Willas		威倫電子股份有限公司 Willas Electronic Corp.		Variant: DRAFT	Git Hash: 34041f4
	Board Name: Motor Control Board			Project Name: Master Hand		
Sheet Title:	File Name: Motor Control - Pinky.kicad_sch	Designer: Damon	Date: 2026-06-27	Revision: + (Unreleased)		
Sheet Path: /Project Architecture/Motor Control - Pinky/		Reviewer:	Size: A4	Sheet: 11 of 23		

[12] Motor Control - Thumb 1



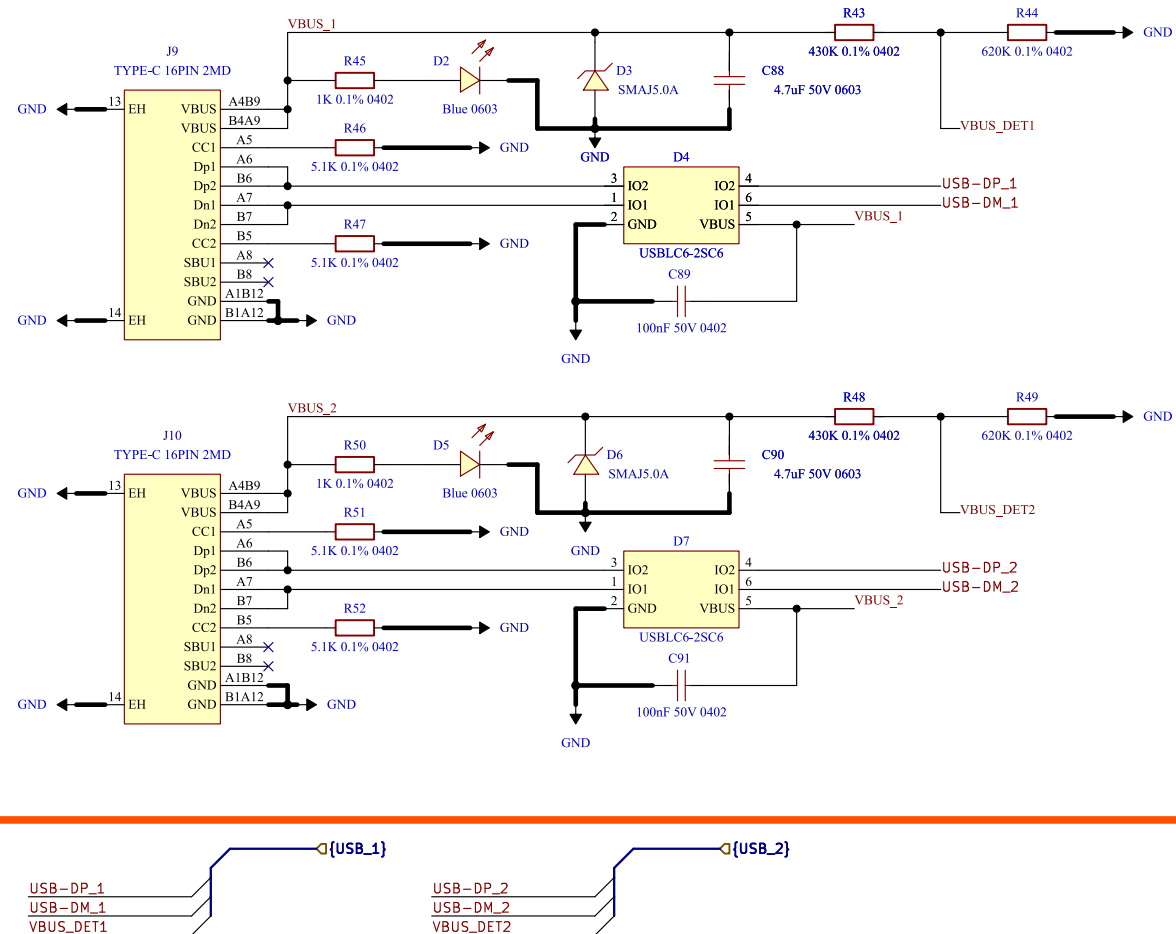
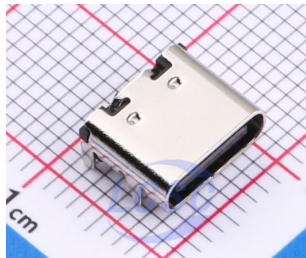
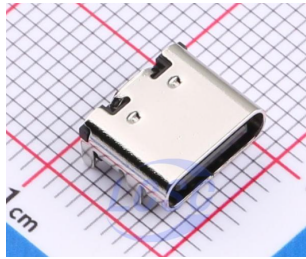
Comments:	Company: 威倫電子股份有限公司 Willas		Variant: DRAFT	Git Hash: 34041f4
	Board Name: Motor Control Board		Project Name: Master Hand	
Sheet Title:	File Name: Motor Control - Thumb 1.kicad_sch	Designer: Damon	Date: 2026-06-27	Revision: + (Unreleased)
Sheet Path: /Project Architecture/Motor Control - Thumb 1/	Reviewer:	Size: A4	Sheet: 12 of 23	

[13] Motor Control - Thumb 2



Comments:	Company: 威倫電子股份有限公司 Willas		Variant: DRAFT	Git Hash: 34041f4
	Board Name: Motor Control Board		Project Name: Master Hand	
Sheet Title:	File Name: Motor Control - Thumb 2.kicad_sch	Designer: Damon	Date: 2026-06-27	Revision: + (Unreleased)
Sheet Path: /Project Architecture/Motor Control - Thumb 2/	Reviewer:	Size: A4	Sheet: 13 of 23	

[14] Interface - USB 2.0



DESIGN NOTE:

- [1] VBUS NOT connected to +5V/+3V3. Board is self-powered(LiPo). Sense + LED only. No backfeed to host.
- [2] LED direct from VBUS @ 1mA. USB2.0 suspend limit 2.5mA. LED 1mA + divider 33uA = 1.03mA (41%). Do NOT raise to 5mA.
- [3] VBUS_DET required - G474 has no dedicated VBUS pin. Self-powered device must not pull up D+ when VBUS=0. FW controls DPPU bit in USB_BCDR. 68k/100k: VBUS 4.40V -> 2.60V > VIH 2.31V. Debounce 100ms.
- [4] D+/D- direct to PA12/PA11. No 22R series. No external 1k5 pull-up (internal RPUI 900-1500R).
- [5] USB clock: PLL-Q 48MHz from HSE 8MHz. ST DFU bootloader uses HS148+CRS - flashable without crystal.
- [6] Both USB ports fully isolated - no shared nets.
- [7] GROUND LOOP WARNING: USB from PC ties host GND to power GND (4.6A peak @20kHz). Use isolator (ADuM3160) or isolated USB-UART when debugging with motors running.

Comments:

Company:

Willas



Variant:

DRAFT

Git Hash:

34041f4

Board Name:

Motor Control Board

Project Name:

Master Hand

Sheet Title:

File Name:

Interface - USB 2.0.kicad_sch

Designer:

Damon

Date:

2026-06-27

Revision:

+(Unreleased)

Sheet Path:

/Project Architecture/Interface - USB 2.0/

Reviewer:

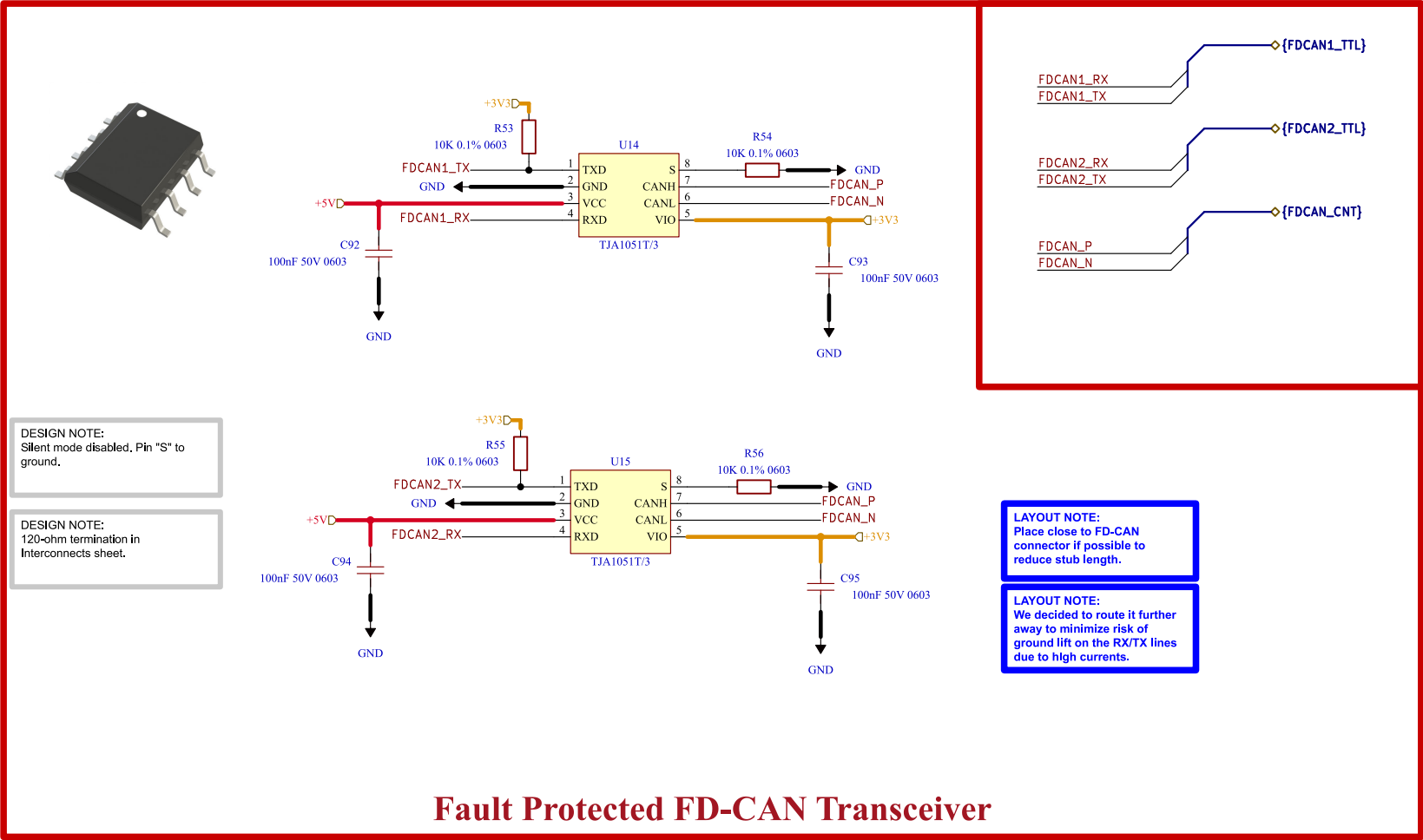
Size:

A4

Sheet:

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[15] Interface - FD-CAN



DESIGN NOTE:
Silent mode disabled, Pin "S" to ground.

DESIGN NOTE:
120-ohm termination in Interconnects sheet.

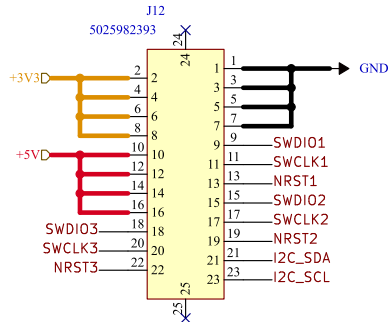
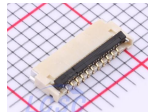
LAYOUT NOTE:
Place close to FD-CAN connector if possible to reduce stub length.

LAYOUT NOTE:
We decided to route it further away to minimize risk of ground lift on the RX/TX lines due to high currents.

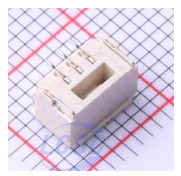
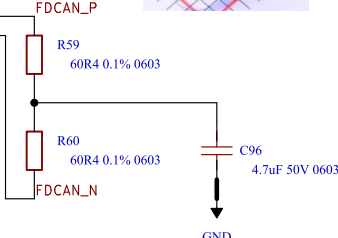
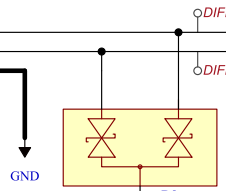
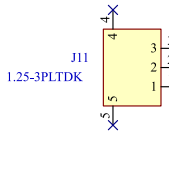
Fault Protected FD-CAN Transceiver

Comments:	Company: Willas Willas Electronic Corp.		Variant: DRAFT		Git Hash: 34041f4
	Board Name: Motor Control Board			Project Name: Master Hand	
Sheet Title:	File Name: Interface - FD-CAN.kicad_sch	Designer: Damon	Date: 2026-06-27	Revision: + (Unreleased)	
Sheet Path: /Project Architecture/Interface - FD-CAN/		Reviewer:	Size: A4	Sheet: 15 of 23	

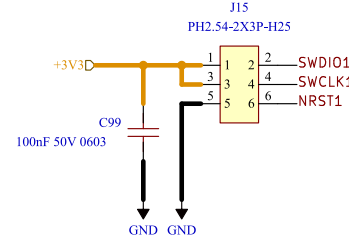
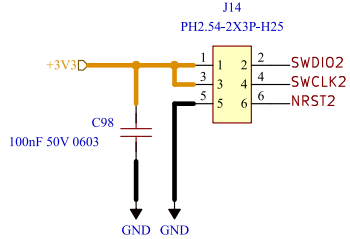
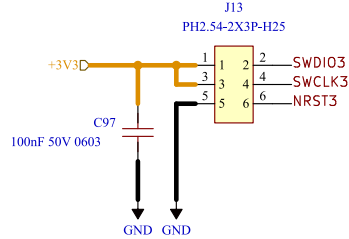
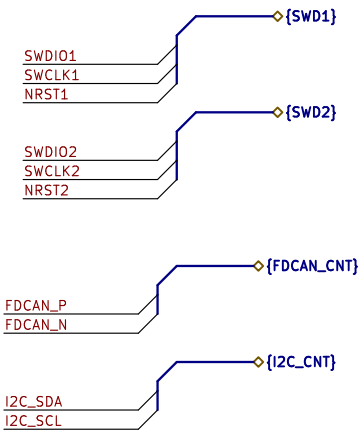
[16] Interface - Interconnects



FFC Connector MCB & MB



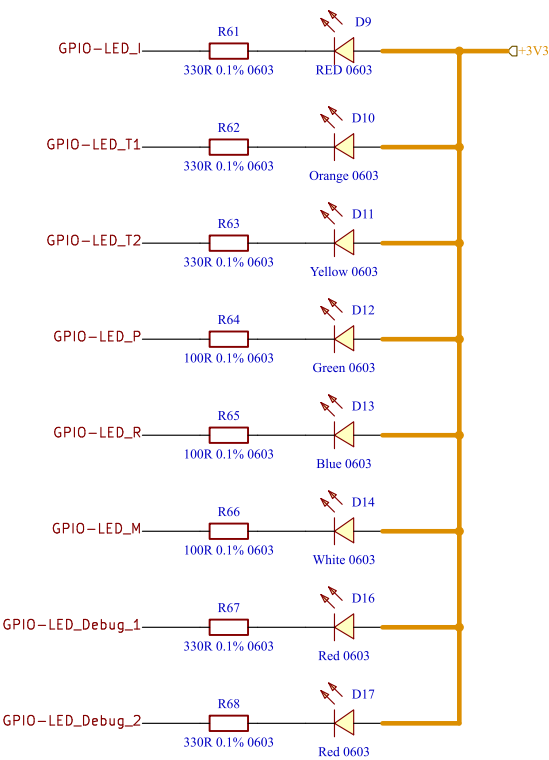
FD-CAN Connector and Termination



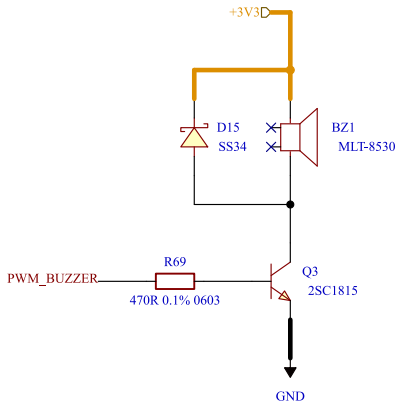
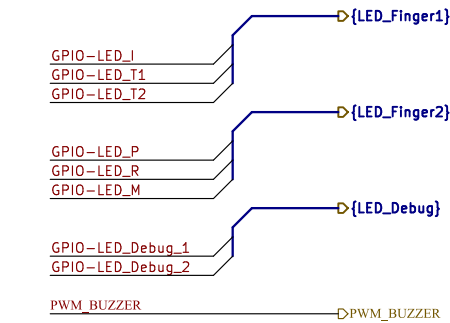
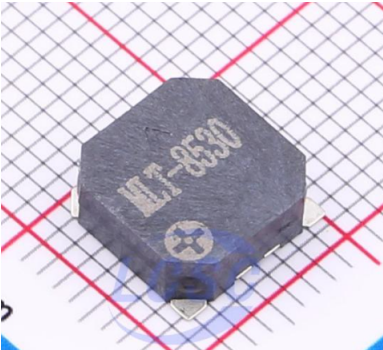
SWD Debug Connector

Comments:	Company: 威倫電子股份有限公司 Willas Willas Electronic Corp.		Variant: DRAFT	Git Hash: 34041f4
	Board Name: Motor Control Board		Project Name: Master Hand	
Sheet Title:	File Name: Interface - Interconnects.kicad_sch	Designer: Damon	Date: 2026-06-27	Revision: + (Unreleased)
Sheet Path: /Project Architecture/Interface - Interconnects/		Reviewer:	Size: A4	Sheet: 16 of 23

[17] User - LED Indicators & Buzzer

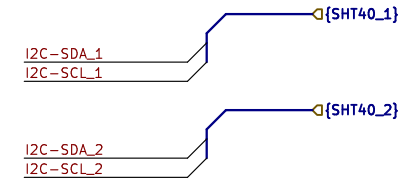
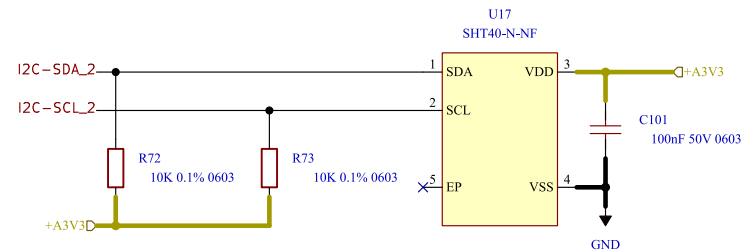
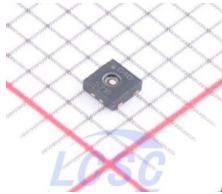
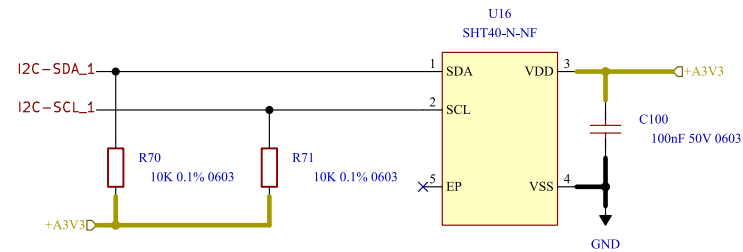
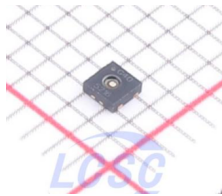


LED Color	Typical Vf Range
Red	1.8 to 2.1
Amber	2 to 2.2
Orange	1.9 to 2.2
Yellow	1.9 to 2.2
Green	2 to 3.1
Blue	3 to 3.7
White	3 to 3.4



Comments:	Company: 威倫電子股份有限公司 Willas		Variant: DRAFT	Git Hash: 34041f4
	Board Name: Motor Control Board		Project Name: Master Hand	
Sheet Title:	File Name: User - LED Indicators & Buzzer.kicad_sch	Designer: Damon	Date: 2026-06-27	Revision: + (Unreleased)
Sheet Path: /Project Architecture/User - LED Indicators & Buzzer/		Reviewer:	Size: A4	Sheet: 17 of 23

[18] Sensing - Temperature & Humidity



Comments:	Company: 威倫電子股份有限公司 Willas		Variant: DRAFT	Git Hash: 34041f4
	Board Name: Motor Control Board		Project Name: Master Hand	
Sheet Title:	File Name: Sensing - Temperature.kicad_sch	Designer: Damon	Date: 2026-06-27	Revision: + (Unreleased)
Sheet Path: /Project Architecture/Sensing - Temperature & Humidity/		Reviewer:	Size: A4	Sheet: 18 of 23

[19] Sensing - Battery

DESIGN NOTE:
Battery voltage sensing divider for 2S LiPo battery.

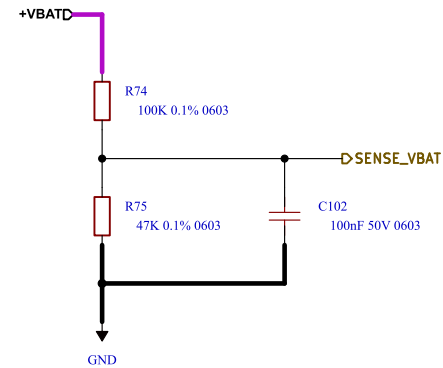
R1 = 100kΩ (Top)
R2 = 47kΩ (Bottom)

Divider Ratio:
 $V_{ADC} = V_{BAT} \times (47 / (100 + 47))$
 $V_{ADC} = V_{BAT} \times 0.3197$

2S LiPo:
Max battery voltage (8.4V): 2.69 V
Nominal voltage (7.4V): 2.37 V
Low battery voltage (6.4V): 2.05 V
Cutoff voltage (6.0V): 1.92 V

Maximum divider current:
 $I = 8.4V / 147k\Omega = 57 \mu A$

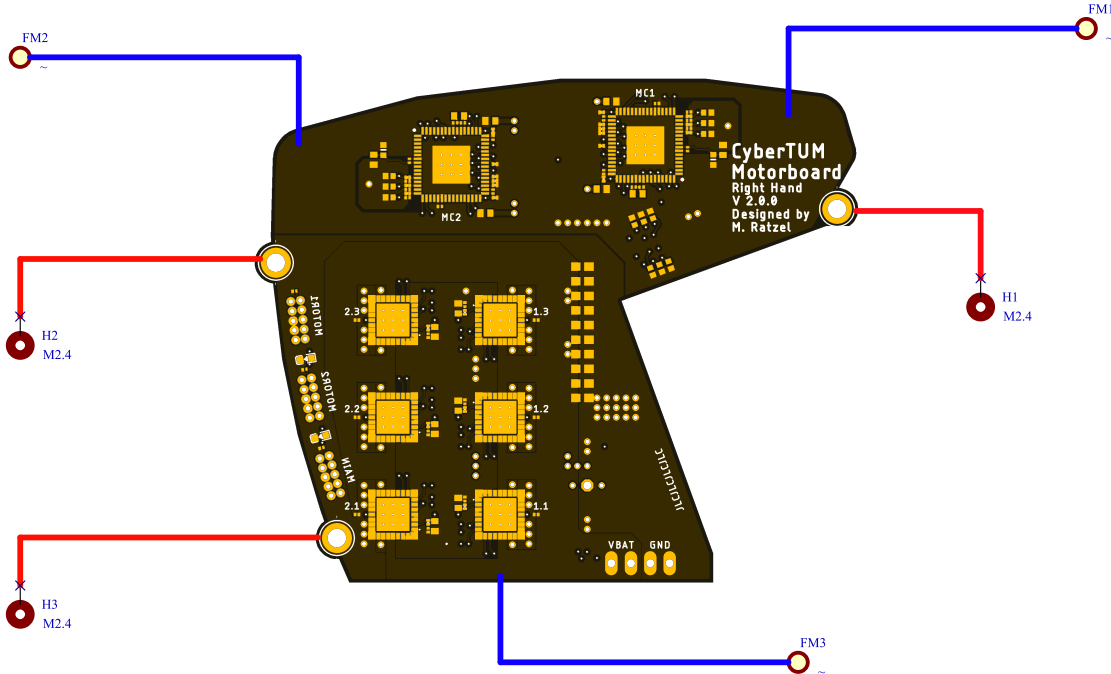
Notes:
- Designed for 3.3V ADC input.
- Add 100nF capacitor at ADC input for noise filtering.
- Use 1% resistors or better.
- Use low temperature coefficient resistors ($\leq 100 \text{ ppm}/^\circ\text{C}$) for improved measurement accuracy.



Battery Voltage Sensing

Comments:	Company: Willas 威倫電子股份有限公司 Willas Electronic Corp.		Variant: DRAFT		Git Hash: 34041f4
	Board Name: Motor Control Board		Project Name: Master Hand		
Sheet Title:	File Name: Sensing - Battery.kicad_sch	Designer: Damon	Date: 2026-06-27	Revision: + (Unreleased)	
Sheet Path: /Project Architecture/Sensing - Battery/		Reviewer:	Size: A4	Sheet: 19 of 23	

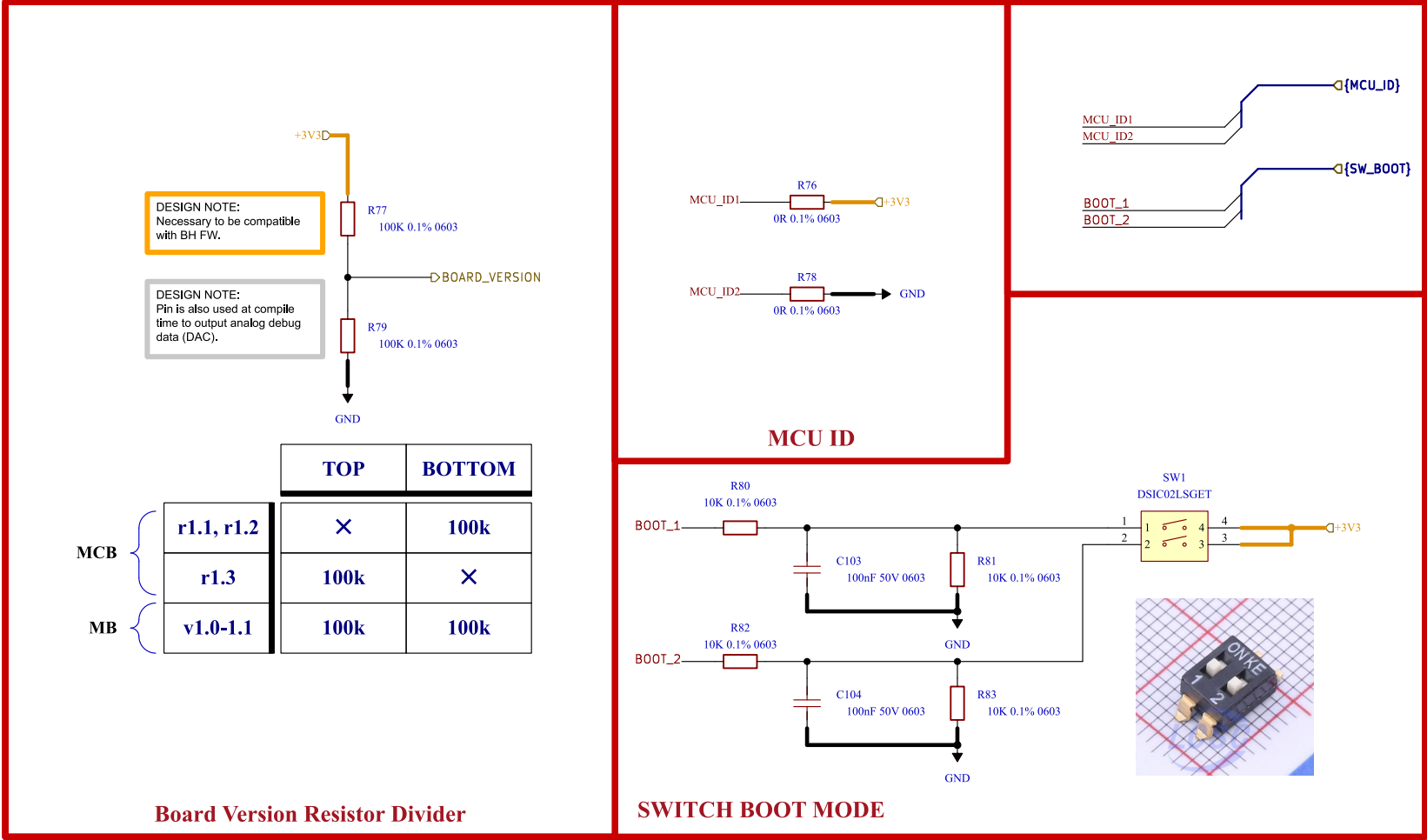
[20] Misc - Holes, Fiducials



Mounting Holes and Fiducial Markers

Comments:	Company: Willas		 威倫電子股份有限公司 Willas Electronic Corp.		Variant: DRAFT	Git Hash: 34041f4
	Board Name: Motor Control Board			Project Name: Master Hand		
Sheet Title:	File Name: Misc - Holes Fiducials.kicad_sch	Designer: Damon	Date: 2026-06-27	Revision: + (Unreleased)		
Sheet Path: /Project Architecture/Misc - Holes, Fiducials/		Reviewer:	Size: A4	Sheet: 20 of 23		

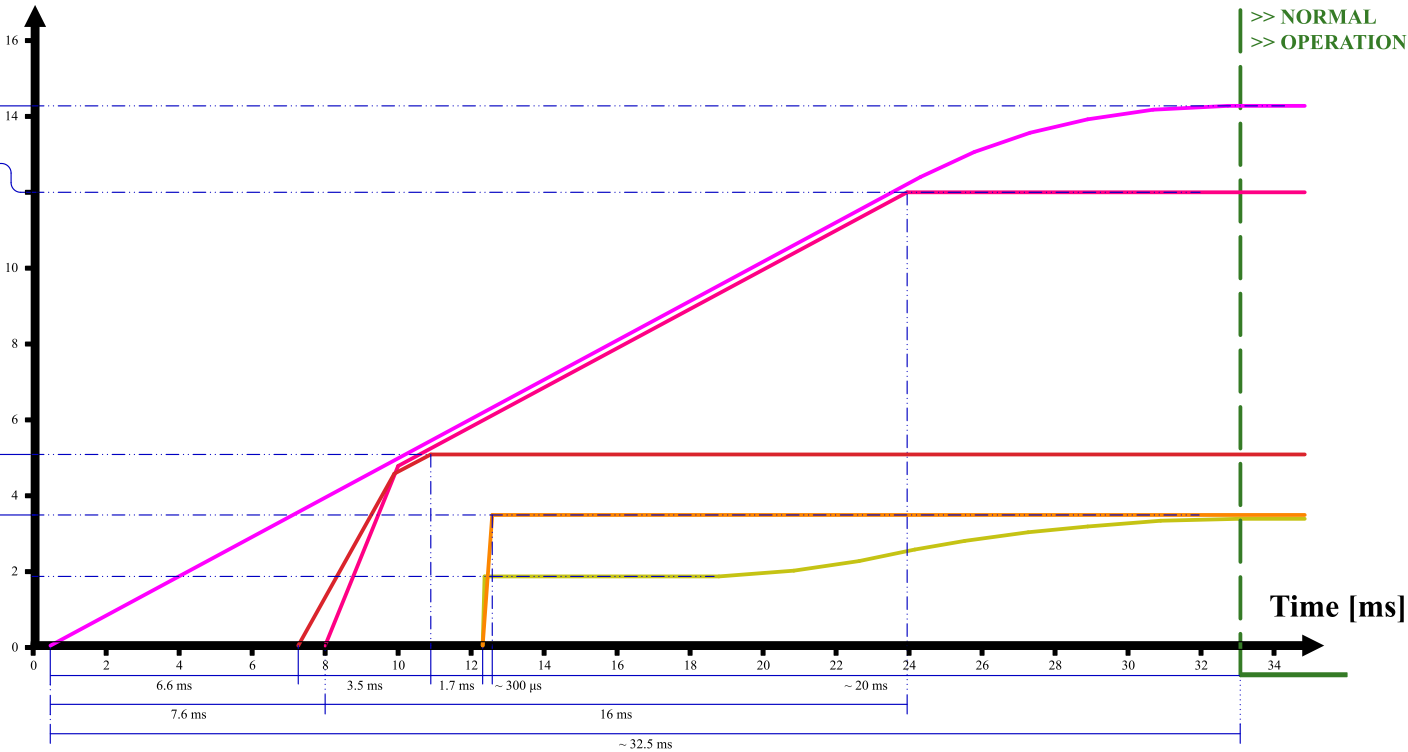
[21] Misc - Board Version, Switch



Comments:	Company: Willas Willas Electronic Corp.		Variant: DRAFT		Git Hash: 34041f4	
	Board Name: Motor Control Board			Project Name: Master Hand		
Sheet Title:	File Name: Misc - Board Version, Switch.kicad_sch	Designer: Damon	Date: 2026-06-27	Revision: + (Unreleased)		
Sheet Path: /Project Architecture/Misc - Board Version, Switch/		Reviewer:	Size: A4	Sheet: 21 of 23		

NAME	SOURCE	LEVEL
+V??	??	? - ? ± ??%
+V??	??	? - ? ± ??%
+V??	??	? - ? ± ??%
+V??	??	? - ? ± ??%
+V??	??	? - ? ± ??%

Voltage [V]



Comments:	Company: Willas		Variant: DRAFT	Git Hash: 34041f4
	Board Name: Motor Control Board		Project Name: Master Hand	
Sheet Title:	File Name: Power - Sequencing.kicad_sch	Designer: Damon	Date: 2026-06-27	Revision: + (Unreleased)
Sheet Path: /Power - Sequencing/		Reviewer:	Size: A4	Sheet: 22 of 23

Comments:	Company: Willas		 威倫電子股份有限公司 Willas Electronic Corp.	Variant: DRAFT	Git Hash: 34041f4
	Board Name: Motor Control Board			Project Name: Master Hand	
Sheet Title:	File Name: Revision History.kicad_sch	Designer: Damon	Date: 2026-06-27	Revision: + (Unreleased)	
Sheet Path: /Revision History/		Reviewer:	Size: A4	Sheet: 23 of 23	